Probability theory and statistics

Lecture Notes

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Ukrainian Catholic University
BA & CS Programmes

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APPLIED SCIENCES FACULTY

Lecture 5. Expectation

Outline

1. Expectation
   - Motivation and definition
   - Properties of expectation
2. Variance
   - Variance and moments
   - Expectation of the product
   - Covariance
3. Generating functions and their applications
   - Definition
   - Examples
   - Properties

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Question

Assume we toss a biased coin n times; how many Heads will we get on average?

- repeat experiment m times;
- count the number X_k of Heads in the kth run;
- the average number of Heads per experiment is

$$\overline{X} := \frac{1}{m}(X_1 + X_2 + \cdots + X_m)$$

Solution

If m is large, we expect that approximately $m \cdot p_X(j)$ of repetitions have j Heads, $j = 0, 1, \ldots, n$. Therefore,

$$X_1 + X_2 + \cdots + X_m \approx \sum_{j=0}^{n} j \cdot m \cdot p_X(j)$$

and

$$\overline{X} \approx \sum_{j=0}^{n} j \cdot p_X(j)$$

Expectation

Definition

Assume that X is a random variable with values $x_1, x_2, \ldots$. If $\sum_k |x_k| p_X(x_k) < \infty$, then the expected value[a] $\mathrm{E}(X)$ of X is

$$\mathrm{E}(X) = \sum_k x_k p_X(x_k)$$

[a] also called the *expectation, mean value, average, or first moment* of X

Example (An indicator **r.v.**)

For any event $A \in \mathscr{F}$, let I_A be the *indicator **r.v.** of the event A*:

$$I_A(\omega) = \begin{cases} 0, & \omega \notin A \\ 1, & \omega \in A \end{cases}$$

The p.m.f. of I_A: $p_I(0) = 1 - \mathrm{P}(A)$, $p_I(1) = \mathrm{P}(A)$, and

$$\mathrm{E}(I_A) = 0 \cdot p_I(0) + 1 \cdot p_I(1) = \mathrm{P}(A)$$

In general, X is an *indicator **r.v.*** if $\mathrm{P}(X = 0) + \mathrm{P}(X = 1) = 1$; then $X = I_A$ for $A := \{X = 1\}$

Example (Binomial distribution)

For X with binomial distribution $\mathscr{B}(n, p)$:

$$\mathrm{E}(X) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k}$$

$$= np(p + 1 - p)^{n-1} = np$$

Example (Infinite expectation)

For X with **p.m.f.** $p_X(k) = 1/[k(k+1)]$, $k \in \mathbb{N}$, the series

$$\sum_{k \geq 1} \frac{k}{k(k+1)} = \sum_{k \geq 1} \frac{1}{k+1}$$

does not converge, thus $\mathrm{E}(X) = +\infty$

Useful formula

If $X \geq 0$ and assumes integer values, then

$$\mathrm{E}(X) = \sum_{k \geq 1} \mathrm{P}(X \geq k)$$

Proof:

$$\sum_{k \geq 1} \mathrm{P}(X \geq k) = \sum_{k \geq 1} \sum_{j \geq k} \mathrm{P}(X = j) = \sum_{j \geq 1} \sum_{k=1}^{j} \mathrm{P}(X = j) = \sum_{j \geq 1} j \, \mathrm{P}(X = j)$$

Example (Geometric distribution)

If $X \sim \mathscr{G}(p)$, then

$$\mathrm{E}(X) = \sum_{k \geq 1} \mathrm{P}(X \geq k) = \sum_{k \geq 1} q^{k-1} = \frac{1}{p}$$

Direct calculations:

$$\begin{aligned} \mathrm{E}(X) &= \textstyle\sum_{k \geq 1} kpq^{k-1} = p\left(\sum_{k \geq 1} q^k\right)' \\ &= p \cdot \left(\frac{1}{1-q}\right)' = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p} \end{aligned}$$

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Functions of **r.v.**

If g is a function on $\mathbb{R}$, what is the distribution of $Y := g \circ X = g(X)$?

- The values of Y are $g(x_k)$;
- label all pairwise distinct values by y_j; then

$$p_Y(y_j) = \mathrm{P}(Y = y_j) = \sum_k \mathrm{P}(g(X) = y_j \mid X = x_k)\mathrm{P}(X = x_k)$$
$$= \sum_{k:g(x_k)=y_j} \mathrm{P}(X = x_k) = \sum_{k:g(x_k)=y_j} p_X(x_k)$$

Theorem (Expectation of $g(X)$)

$$\boxed{\mathrm{E}(g(X)) = \sum_k g(x_k)p_X(x_k)}$$

Indeed, $$\mathrm{E}(Y) = \sum_j y_j\, p_Y(y_j) = \sum_j y_j \sum_{k:g(x_k)=y_j} p_X(x_k)$$
$$= \sum_j \sum_{k:g(x_k)=y_j} g(x_k)p_X(x_k) = \sum_k g(x_k)p_X(x_k)$$

Example (St. Petersburg paradox)

- T is the no. of games played; $\mathrm{P}(T = k) = q^{k-1}p$, $k \in \mathbb{N}$
- T is finite since $\sum_{k\geq 1} \mathrm{P}(T = k) = 1$; thus we always win 1
- X is the winning bet; $X = 2^{T-1}$
- mean value of the winning bet:

$$\mathrm{E}(X) = \mathrm{E}(2^{T-1}) = \sum_{k\geq 1} 2^{k-1}q^{k-1}p = +\infty$$

  since $2q \geq 1$
- if at most N games can be played, then the expected gain is

$$\mathrm{P}\{T \leq N\} - (2^N - 1)\mathrm{P}\{T > N\} = 1 - 2^N\mathrm{P}\{T > N\} = 1 - (2q)^N \leq 0$$

Linearity of expectation

Theorem: Assume X and Y are **r.v.** with joint **p.m.f.** $p_{X,Y}$; then

$$\boxed{\mathrm{E}(aX + bY) = a\mathrm{E}(X) + b\mathrm{E}(Y)}$$

Remark: An analogue of the previous thm reads

$$\mathrm{E}(g(X, Y)) = \sum_{k,j} g(x_k, y_j) p_{X,Y}(x_k, y_j)$$

Proof.

$$\begin{aligned}
\mathrm{E}(aX + bY) &= \sum_{k,j} (ax_k + by_j) p_{X,Y}(x_k, y_j) \\
&= \sum_{k,j} ax_k p_{X,Y}(x_k, y_j) + \sum_{k,j} by_j p_{X,Y}(x_k, y_j) \\
&= a \sum_k x_k \sum_j p_{X,Y}(x_k, y_j) + b \sum_j y_j \sum_k p_{X,Y}(x_k, y_j) \\
&= a \sum_k x_k p_X(x_k) + b \sum_j y_j p_Y(y_j) = a\mathrm{E}(X) + b\mathrm{E}(Y)
\end{aligned}$$

Randomized quicksort algorithm

Aim is to sort any set of numbers in increasing order

Example: $2, 7, 6, \boxed{3}, 1, 8, 5, 4$

$$\underbrace{2, 1}_{<3}, \boxed{3}, \underbrace{6, 7, 8, 5, 4}_{>3}$$

Then recursion on both parts: e.g., $6, 7, \boxed{8}, 5, 4 \quad \mapsto \quad \underbrace{6, 7, 5, 4}_{<8}, \boxed{8}$

then continue with $6, 7, \boxed{5}, 4 \quad \mapsto \quad 4, \boxed{5}, 7, 6 \quad \mapsto \quad 4, 5, 6, 7$

Total number of comparisons: $7 + 1 + 4 + 3 + 1 = 16$;
worst case: $O(n^2)$

Idea: randomised pivoting, i.e., on each step, choose a pivot randomly

Question: how many comparisons on average?
Answer: $O(n \log n)$

Randomized quicksort algorithm

Input: any of $n!$ equally likely permutations of $1, 2, \ldots, n$

- X is a random number of comparisons made when pivots are chosen uniformly at random
- were i and j ($i < j$) compared?
- X_{ij} a comparison indicator ($= 1$ if i and j got compared, $= 0$ else)
- $X_{ij} = 1$ iff i or j was a pivot before any of $i + 1, \ldots, j - 1$ (otherwise i and j are moved to different parts sorted independently)
- thus $\mathrm{E}(X_{ij}) = \mathrm{P}(X_{ij} = 1) = 2/(j - i + 1)$ and

$$\mathrm{E}(X) = \sum_{i<j} \mathrm{E}(X_{ij}) = \sum_{i<j} \frac{2}{j-i+1} = \sum_{i=1}^{n} \sum_{j=i+1}^{n} \frac{2}{j-i+1}$$
$$\leq \sum_{i=1}^{n} \sum_{k=1}^{n} \frac{2}{k} = 2n \sum_{k=1}^{n} \frac{1}{k} = O(n \log n)$$

Inclusion-exclusion formula: For arbitrary $A_1, \ldots, A_n$,

$$\mathrm{P}(A_1 \cup \cdots \cup A_n) = \sum_k \mathrm{P}(A_k) - \sum_{1 \le i_1 < i_2 \le n} \mathrm{P}(A_{i_1} A_{i_2}) + \sum_{1 \le i_1 < i_2 < i_3 \le n} \mathrm{P}(A_{i_1} A_{i_2} A_{i_3}) + \cdots + (-)^{n-1} \mathrm{P}(A_1 \cdots A_n)$$

Proof.

- Set $A = \cup A_k$; by de Morgan's rule $A^c = A_1^c \cdots A_n^c$
- In terms of indicators, $I_{A^c} = I_{A_1^c} \cdots I_{A_n^c}$, i.e.,

$$1 - I_A = \prod (1 - I_{A_k}) = 1 - \sum I_{A_k} + \sum_{1 \le i_1 < i_2 \le n} I_{A_{i_1}} I_{A_{i_2}} + \cdots$$
$$= 1 - \sum I_{A_k} + \sum_{1 \le i_1 < i_2 \le n} I_{A_{i_1} \cap A_{i_2}} + \cdots$$

- take expectations and recall that $\mathrm{E}(I_B) = \mathrm{P}(B)$

Total probability rule for expectations

Total probability rule for expectations

If $B_1, \ldots, B_n$ is a partition of the sample space, then

$$\boxed{\mathrm{E}(X) = \mathrm{E}(X \mid B_1)\mathrm{P}(B_1) + \cdots + \mathrm{E}(X \mid B_n)\mathrm{P}(B_n)}$$

Reason: for every x, $\quad p_x(x) = \mathrm{P}(X = x) = \sum_j \mathrm{P}(X = x \mid B_j)\mathrm{P}(B_j)$

Example

Let N be the number of tosses of a non-symmetric coin until the first occurrence of two Heads in row. Then

$$\mathrm{E}(N) = \mathrm{E}(N \mid T_1)\mathrm{P}(T_1) + \mathrm{E}(N \mid H_1 T_2)\mathrm{P}(H_1 T_2) + \mathrm{E}(N \mid H_1 H_2)\mathrm{P}(H_1 H_2)$$

If $a := \mathrm{E}(N)$, then $a = (a+1)q + (a+2)pq + 2p^2$, so that

$$\mathrm{E}(N) = \frac{q + 2pq + 2p^2}{1 - q - pq} = \frac{q + 2p}{p^2} = \frac{1}{p} + \frac{1}{p^2}$$

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Variance and moments

Definition (Variance and moments)

Assume the **p.m.f.** p_X of a **r.v.** X satisfies $\sum |x|^k p_X(x) < \infty$. Then

- the kth moment of X is $\mu_k = \mathrm{E}(X^k)$;
- the kth central moment of X is $\sigma_k = \mathrm{E}\big((X - \mathrm{E}(X))^k\big)$;
- the kth factorial moment of X is $\mu^{(k)} = \mathrm{E}(X(X-1)\cdots(X-k+1))$
- σ_2 is called the variance of X and is denoted σ^2 or $\mathrm{Var}(X)$:

$$\mathrm{Var}(X) = \mathrm{E}\big((X - \mathrm{E}(X))^2\big)$$

  Variance is a measure of dispersion, or spread around the mean
- The standard deviation of X is $\sigma_X = \sqrt{\mathrm{Var}(X)}$

Example

For the indicator I, $I - \mathrm{E}(I)$ assumes values $1-p$ and $-p$ with probabilities p and $1-p$ respectively, thus

$$\sigma_I^2 = \mathrm{E}\big((I - \mathrm{E}(I))^2\big) = (1-p)^2 p + (-p)^2(1-p) = p(1-p)$$

Example

Properties of variance: alternative formula and scaling

$$\begin{aligned}
\mathrm{E}\big((X-\mathrm{E}(X))^2\big) &= \mathrm{E}(X^2 - 2\mathrm{E}(X)X + \mathrm{E}(X)^2) \\
&= \mathrm{E}(X^2) - 2\mathrm{E}(X)\mathrm{E}(X) + \mathrm{E}(X)^2 = \mathrm{E}(X^2) - \mathrm{E}(X)^2 \\
\mathrm{Var}(aX) &= \mathrm{E}\big((aX - \mathrm{E}(aX))^2\big) = \mathrm{E}\big(a^2(X-\mathrm{E}(X))^2\big) \\
&= a^2\mathrm{E}\big((X-\mathrm{E}(X))^2\big) = a^2\mathrm{Var}(X)
\end{aligned}$$

The binomial distribution

- Probabilistic meaning of $X \sim \mathscr{B}(n,p)$: numbers of successes in the Bernoulli trials (e.g., number of Heads in n tossing of a biased coin)
- denote by I_k the success indicator of the kth trial: $I_k = 1$ if that trial was a success and $I_k = 0$ otherwise
- then $X = I_1 + I_2 + \cdots + I_n$
- Therefore, $\mathrm{E}(X) = \mathrm{E}(I_1) + \mathrm{E}(I_2) + \cdots + \mathrm{E}(I_n) = n\mathrm{E}(I_1) = np$

Binomial distribution

Example

The 2nd factorial moment of $X \sim \mathscr{B}(n,p)$ is

$$\mu^{(2)}(X) = \mathrm{E}(X(X-1)) = \sum_{k=0}^{n} k(k-1)p_X(k)$$

$$= n(n-1)p^2 \sum_{j=0}^{n-2} \binom{n-2}{j} p^j (1-p)^{n-k} = n(n-1)p^2$$

Therefore,

$$\mathrm{Var}(X) = \sigma_X^2 = \mathrm{E}(X^2 - X + X) - \mathrm{E}(X)^2 = \mu^{(2)}(X) + \mathrm{E}(X) - \mathrm{E}(X)^2$$

$$= n(n-1)p^2 + np - n^2p^2 = np - np^2 = np(1-p)$$

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Expectation of the product

Theorem: If X and Y are independent, then

$$\boxed{\mathrm{E}(X \cdot Y) = \mathrm{E}(X) \cdot \mathrm{E}(Y)}$$

Proof.

$$\begin{aligned}\mathrm{E}(X \cdot Y) &= \sum\nolimits_{k,j} (x_k \cdot y_j)\, p_{X,Y}(x_k, y_j) \\ &= \sum\nolimits_{k,j} (x_k \cdot y_j)\, p_X(x_k) p_Y(y_j) \\ &= \sum\nolimits_k x_k p_X(x_k) \cdot \sum\nolimits_j y_j p_Y(y_j) = \mathrm{E}(X) \cdot \mathrm{E}(Y)\end{aligned}$$

□

Example (Indicators I_k and I_j in the Bernoulli experiment)

$$\mathrm{E}(I_k I_j) = \mathrm{E}(I_k) \cdot \mathrm{E}(I_j) = p^2 \qquad \text{if} \qquad k \neq j;$$

thus for $X = I_1 + \cdots + I_n \sim \mathscr{B}(n, p)$ we find that

$$\mathrm{E}(X^2) = \mathrm{E}((I_1 + \cdots + I_n)^2) = \sum \mathrm{E}(I_k^2) + \sum\nolimits_{k \neq j} \mathrm{E}(I_k I_j) = np + n(n-1)p^2$$

and $\mu^{(2)}(X) = \mathrm{E}(X^2) - \mathrm{E}(X) = n(n-1)p^2$ as found above

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Covariance

Definition

The covariance of **r.v.** X and Y is

$$\mathrm{Cov}(X, Y) = \mathrm{E}(X - \mathrm{E}(X))(Y - \mathrm{E}(Y)) = \mathrm{E}(XY) - \mathrm{E}(X)\mathrm{E}(Y)$$

Corollary

If X and Y are independent, then $\mathrm{Cov}(X, Y) = 0$.

Corollary

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X, Y) + \mathrm{Var}(Y)$$

Proof.

$\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$, so that

$$(X + Y) - \mathrm{E}(X + Y) = (X - \mathrm{E}(X)) + (Y - \mathrm{E}(Y)),$$

and

$$\begin{aligned}\mathrm{Var}(X + Y) &= \mathrm{E}[(X - \mathrm{E}(X)) + (Y - \mathrm{E}(Y))]^2 \\ &= \mathrm{Var}(X) + 2\mathrm{Cov}(X, Y) + \mathrm{Var}(Y)\end{aligned}$$

Variance of the sum

Theorem: If X and Y are independent, then

$$\boxed{\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y)}$$

Proof.

By the above,

$$\mathrm{Var}(aX + bY) = \mathrm{Var}(aX) + 2ab\mathrm{Cov}(X, Y) + \mathrm{Var}(bY)$$

□

Example (Binomial distribution)

Recall that $X \sim \mathscr{B}(n, p)$ is equal to $I_1 + I_2 + \ldots I_n$, there I_k is the indicator of success in k^{th} trial of the Bernoulli experiment. As the indicators are independent,

$$\mathrm{Var}(X) = \mathrm{Var}(I_1 + \cdots + I_n) = \mathrm{Var}(I_1) + \cdots + \mathrm{Var}(I_n) = n\mathrm{Var}(I_1) = np(1-p)$$

Covariance vs correlation coefficient

Meaning of covariance: $\mathrm{Cov}(X, Y) > 0 \implies$

- X and Y dependent
- larger values of X *typically* correspond to larger values of Y

(positive correlation)

Remark

- $\mathrm{Cov}(X, Y) = 0$ *does not mean X and Y are independent*
- $\mathrm{Cov}(X, Y)$ *is a bilinear form*

Correlation coefficient between X and Y:

$$\boxed{\mathrm{Cor}(X, Y) = \rho(X, Y) := \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\sigma(Y)}}$$

Properties of $\mathrm{Cor}(X, Y)$

- $|\mathrm{Cor}(X, Y)| \leq 1$ (Cauchy–Bunyakowsky–Schwarz inequality)
- $\mathrm{Cor}(X, Y) = \pm 1$ implies $Y = aX + b$

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Probability generating functions

Definition

The probability generating function G_X of the integer valued **r.v.** X is defined by

$$G_X(s) = \mathrm{E}(s^X) = \sum_k p_X(k)s^k, \qquad s \in \mathbb{R}$$

Example

For the indicator **r.v.** I, $G_I(s) = (1-p)s^0 + ps^1 = 1 - p + ps$

Properties of **p.g.f**:

- $p_X(k) = G^{(k)}(0)/k!$ (so G generates **p.m.f.**)
- $G_X^{(k)}(1) = \mu^{(k)}$ (the kth factorial moment of X)
- If $Y = X + a$, then $G_Y(s) = s^a G_X(s)$
- If X and Y are independent, then $G_{X+Y}(s) = G_X(s)G_Y(s)$

The latter property follows from the fact that s^X and s^Y are independent **r.v.**, so that

$$G_{X+Y}(s) = \mathrm{E}(s^X s^Y) = \mathrm{E}(s^X)\mathrm{E}(s^Y) = G_X(s)G_Y(s)$$

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Examples

Example (Binomial distribution)

$$X = I_1 + \cdots + I_n \implies G_X(s) = G_{I_1}(s) \cdots G_{I_n}(s) = \left(G_{I_1}(s)\right)^n = (1-p+ps)^n$$

Example (Sum of independent binomial distributions)

If $X \sim \mathscr{B}(n,p)$ and $Y \sim \mathscr{B}(m,p)$ are independent,

$$G_{X+Y}(s) = G_X(s)G_Y(s) = (1-p+ps)^{n+m}$$

Example (Poisson distribution $X \sim \mathscr{P}(\lambda)$)

$$G_X(s) = \sum_{k\geq 0} \frac{\lambda^k}{k!} e^{-\lambda} s^k = e^{\lambda(s-1)}$$

Therefore, $\mu_X^{(k)} = \lambda^k$

Example (Sum of independent Poisson distributions)

If $X \sim \mathscr{P}(\lambda_1)$ and $Y \sim \mathscr{P}(\lambda_2)$ are independent,

$$G_{X+Y}(s) = G_X(s)G_Y(s) = e^{(\lambda_1+\lambda_2)(s-1)}$$

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Uniqueness

Theorem (Uniqueness)

Let X and Y have generating functions $G_X(s)$ and $G_Y(s)$. If

$$G_X(s) = G_Y(s) \quad \text{for} \quad |s| < 1,$$

then X and Y have the same probability mass function.

Proof.

Follows from uniqueness of the power series $\sum_k p_X(k)s^k$ □

Corollary

$X \sim \mathscr{B}(n,p)$ and $Y \sim \mathscr{B}(m,p)$ are independent, then
$X + Y \sim \mathscr{B}(n+m,p)$

Corollary

If $X \sim \mathscr{P}(\lambda_1)$ and $Y \sim \mathscr{P}(\lambda_2)$ are independent, then
$X + Y \sim \mathscr{P}(\lambda_1 + \lambda_2)$

Example: Let N be the number of tosses of a coin until two Heads occur in row

By the total probability rule for expectations, $G_N(s) = \mathrm{E}(s^N)$ satisfies

$$\begin{aligned} G_N(s) &= \mathrm{E}(s^N \mid T_1)\mathrm{P}(T_1) + \mathrm{E}(s^N \mid H_1T_2)\mathrm{P}(H_1T_2) + \mathrm{E}(s^N \mid H_1H_2)\mathrm{P}(H_1H_2) \\ &= \mathrm{E}(s^{N+1})q + \mathrm{E}(s^{N+2})pq + s^2p^2 \end{aligned}$$

so that

$$\boxed{G_N(s) = \frac{s^2p^2}{1 - sq - s^2pq}}$$

In particular,

$$\begin{aligned} \mathrm{E}(N) = G'_N(1) &= \frac{2sp^2(1 - sq - s^2pq) + s^2p^2(q + 2spq)}{(1 - sq - s^2pq)^2}\Big|_{s=1} \\ &= \frac{2p^2 + q + 2pq}{p^2} = \frac{1 + p}{p^2} \end{aligned}$$

As explained above, $G_N^{(k)}(0)/k! = p_N(k)$, which gives the **p.m.f.** of N